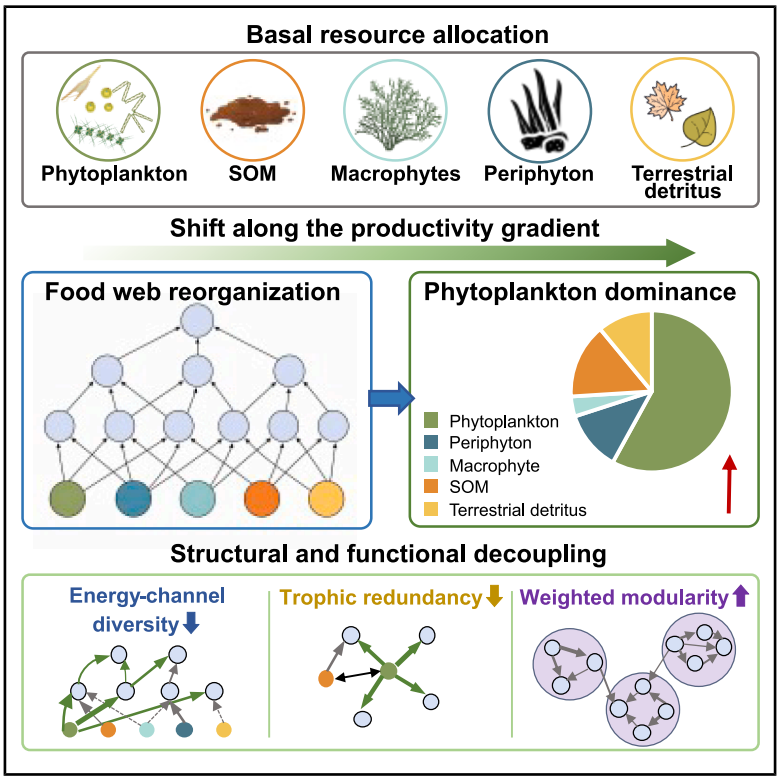
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Article

Phytoplankton dominance alters basal energy pathways and food web structure in eutrophic shallow lakes

Graphical abstract



Highlights

- Eutrophication drives basal energy allocation toward a phytoplankton-based pathway
- Phytoplankton dominance reduces energy-channel diversity and trophic redundancy
- Food webs become more modular as phytoplankton dominance increases
- Eutrophication drives structural and functional decoupling in food webs

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In brief

Earth sciences; environmental science; aquatic science

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Phytoplankton dominance alters basal energy pathways and food web structure in eutrophic shallow lakes

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SUMMARY

Eutrophication is widely recognized for increasing primary production, yet how changes in basal energy allocation reorganize food web architecture and stability remains insufficiently resolved. Across three eutrophic shallow lakes spanning a productivity gradient, we combined stable isotope analysis with a Bayesian mixing model (EcoDiet) to quantify the relative contributions of five basal resources to consumers. We then linked phytoplankton dominance to energy-channel diversity, trophic redundancy, and interaction-weighted modularity. Results showed that increasing phytoplankton dominance progressively concentrated basal carbon allocation into a single pelagic pathway. This shift reduced energy-channel diversity and trophic redundancy, suggesting erosion of functional buffering capacity. Conversely, phytoplankton dominance simultaneously strengthened network modularity. These results revealed a structural-functional decoupling that shifted food webs toward more pelagic-based trophic pathways and structurally modular configurations. Thus, nutrient enrichment influences ecosystem stability in multidimensional ways, altering not only productivity but also the distribution and pathways of energy through trophic networks.

INTRODUCTION

Eutrophication is one of the most widespread and persistent stressors affecting aquatic ecosystems worldwide.[1,2] It is extensively accelerated by anthropogenic activities like urban wastewater discharge, agricultural runoff, and land-use change.[3] Shallow lakes are especially susceptible to nutrient enrichment due to their shallow water columns, strong benthic-pelagic coupling, and high sensitivity to external nutrient inputs. As intensifying eutrophication, these lakes often experience a cascade of ecological changes, including increased primary production[4] and shifts in community composition.[5–7] These changes could induce a transition to an alternative stable state from a clear-water regime with submerged macrophytes to a turbid-water state dominated by phytoplankton,[8–10] which ultimately alters food web structure and ecosystem functioning.[11–13]

Generally, eutrophication has been evaluated mainly through nutrient loading and variations in primary production.[14,15] However, there is accumulating evidence that food web stability depends not only on the amounts of available resources, but also on the way carbon sources are allocated at the base of food webs and subsequently conveyed through trophic pathways.[16,17] In shallow lakes, basal production is supported by multiple coexisting pathways, including phytoplankton, benthic algae, macrophytes, and autochthonous and allochthonous organic matter. The relative contributions of these sources, which vary across ecosystems and environmental conditions, influence consumer diet composition, trophic interactions and the overall food web structure.[18–20] Increased nutrient loading catalyzes a rapid shift toward phytoplankton dominance, thereby reshaping basal energy pathways and disrupting the balance among alternative energy channels.[21,22]

Phytoplankton-dominated food webs typically develop strengthened pelagic energy channels and relatively short trophic transfer distances, which can facilitate the efficiency of energy transfer to higher trophic levels.[23,24] Such reliance on a singular basal source, while energetically advantageous, tends to erode the diversity of energy pathways and trophic redundancy,

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Table 1. Variations in physicochemical parameters and biomass across the three plateau lakes

Parameters	Yilong lake	Xingyun lake	Qilu lake	*p* value
Physicochemical parameters				
Water temperature (°C)	24.13 ± 0.31 **a**	21.90 ± 0.67 **b**	22.83 ± 0.92 **b**	**<0.001**
pH	8.64 ± 0.03 **b**	9.88 ± 0.07 **a**	9.55 ± 0.43 **a**	**<0.001**
Conductivity (μs/cm)	630.00 ± 6.36 **c**	657.33 ± 7.76 **b**	828.33 ± 15.34 **a**	**<0.001**
Chlorophyll-sa (μg/L)	9.24 ± 3.32 **c**	29.08 ± 4.68 **b**	87.74 ± 14.09 **a**	**<0.001**
Total phosphorus (mg/L)	0.02 ± 0.01 **c**	0.05 ± 0.02 **b**	0.07 ± 0.01 **a**	**<0.001**
Total nitrogen (mg/L)	1.61 ± 0.22 **b**	1.36 ± 0.19 **b**	2.07 ± 0.20 **a**	**<0.001**
Dissolved oxygen (mg/L)	6.25 ± 1.06	5.56 ± 0.66	6.16 ± 1.50	0.524
Biomass				
Phytoplankton (mg/L)	3.02 ± 0.69 **b**	2.66 ± 1.09 **b**	8.77 ± 0.91 **a**	**<0.001**
Zooplankton (mg/L)	2.15 ± 0.64 **a**	0.09 ± 0.04 **b**	0.66 ± 0.34 **b**	**<0.001**
Zoobenthos (g/m^2)	0.91 ± 0.55 **b**	13.12 ± 7.22 **a**	0.13 ± 0.12 **c**	**<0.001**

Bold *p* values indicate significant differences in the one-way ANOVA results ($p < 0.05$), and bold lowercase letters following the values denote the results of post hoc multiple comparison tests (Tukey HSD).

compromising functional buffering capacity.[17] A ecological trade-off thus emerges. Gains in short-term productivity and transfer efficiency appear to come at the expense of long-term structural resilience and resistance since the network becomes more sensitive to external perturbations.[25] Apart from the simplified energy pathway, nutrient loading-induced food web reorganization has generally resulted in structural changes, e.g., compartmentalization, that influences the disturbance propagation pathways in the ecological network.[26] Despite the growing appreciation of the role of basal resource structure in ecosystem functioning,[16,27,28] empirical evidence linking consumer resource use and food web stability remains scanty, particularly in the dynamics of shallow plateau lakes.

Across three eutrophic shallow lakes that represent a productivity gradient, we investigated how the rise of phytoplankton dominance reconfigures basal energy pathways and food web architecture. By coupling stable isotope analysis with a Bayesian dietary mixing model (EcoDiet), the relative contributions of different food sources to consumers and the relative importance of five basal resources (phytoplankton, periphyton, submerged macrophytes, sedimentary organic matter [SOM], and terrestrial detritus) within the food web were estimated. Subsequently, we adopted a structural, multi-dimensional perspective to evaluate the relationship between phytoplankton dominance and a suite of complementary stability-related metrics, including basal energy-channel diversity, trophic redundancy, and interaction-weighted modularity. We assessed the proposition that eutrophication increases consumer reliance on phytoplankton-derived energy, consequently reducing energy-channel diversity and trophic redundancy. Additionally, we hypothesized that phytoplankton-dominated food webs are more modular, representing a more compartmentalized structure that may restrict the propagation of localized disturbances. By integrating basal resources utilization with food web structure and stability, this study aims to advance the understanding of how eutrophication alters shallow lake food webs and to inform the management of eutrophic freshwater ecosystems.

RESULTS

Water quality characteristics and biological community structure

Significant inter-lake variations in physicochemical conditions and community structure were detected among the three plateau lakes (Table 1). Although all lakes were characterized as eutrophic systems, they exhibited pronounced differences in trophic intensity and primary productivity. Chlorophyll-a concentrations increased markedly from Yilong Lake (9.24 ± 3.32 μg L^{-1}) to Xingyun Lake (29.08 ± 4.68 μg L^{-1}) and were highest in Qilu Lake (87.74 ± 14.09 μg L^{-1}). Total phosphorus (TP) showed a 3.5-fold increase across lakes (0.02–0.07 mg L^{-1}), while total nitrogen (TN) reached its highest concentration in Qilu Lake (2.07 ± 0.20 mg L^{-1}). Phytoplankton biomass exhibited a similar pattern, with significantly greater biomass in Qilu Lake compared to the other two lakes. Moreover, phytoplankton biomass was positively correlated with TN and TP (Figure S1), highlighting nutrient availability acting as a key driver of primary production. These results indicated a clear gradient of increasing phytoplankton dominance and trophic intensity across the three eutrophic lakes.

In addition, Yilong Lake exhibited the highest surface water temperature while maintaining the lowest pH among the three lakes. Conductivity increased from Yilong Lake to Xingyun Lake, reaching its maximum in Qilu Lake. Dissolved oxygen displayed no significant differences among the three lakes ($p = 0.524$). Community biomass patterns also varied markedly. Zooplankton biomass peaked in Yilong Lake, whereas zoobenthos biomass was highest in Xingyun Lake and lowest in Qilu Lake.

Stable isotope signatures and food web structure

The $\delta^{13}C$ values of all consumers were fully encompassed within the isotopic ranges of sampled basal sources, suggesting that these sources represented the potential carbon inputs to the food webs (Figure 1A; Table S1). Comparison of dual-isotope ($\delta^{13}C$-$\delta^{15}N$) biplots among lakes revealed a consistent

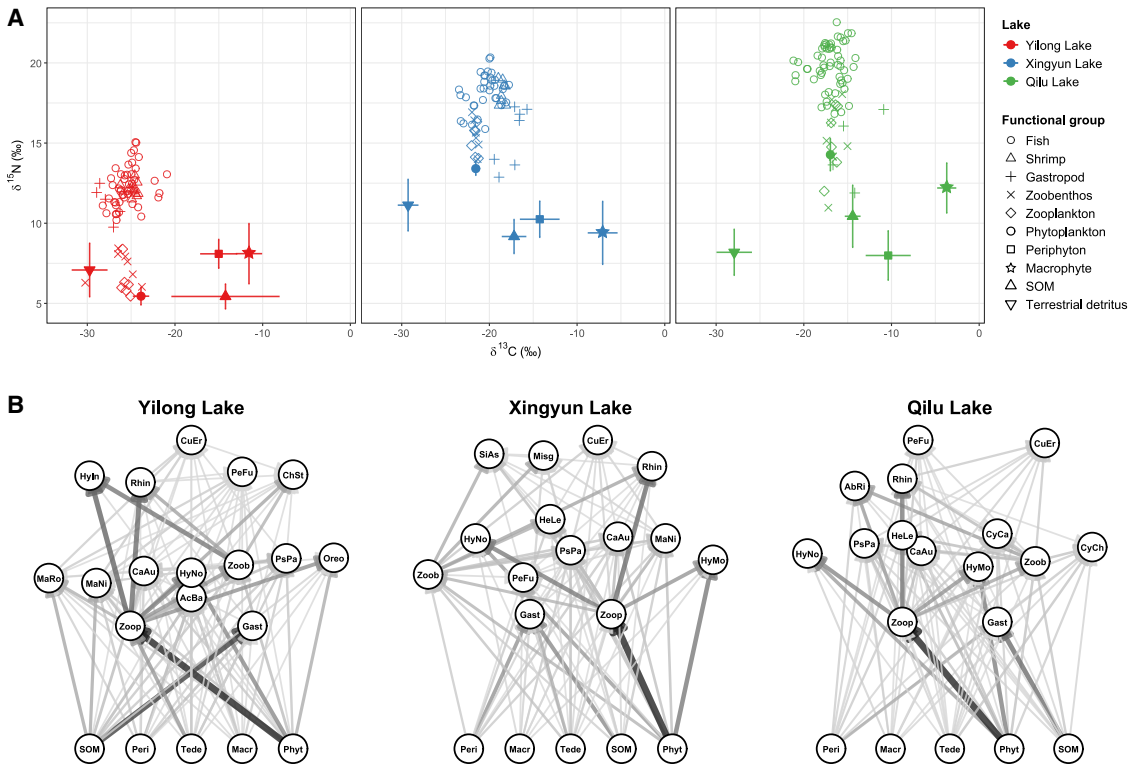


Figure 1. Carbon and nitrogen stable isotope signatures and food web structure in the three lakes

(A) Carbon and nitrogen stable isotope signatures and (B) food web structure in the three lakes.

The thickness of the lines represents the posterior median of dietary contribution proportions. SiAs, *Silurus asotus*; CuEr, *Culter erythropterus*; ChSt, *Channa striata*; HyIn, *Hyporhamphus intermedius*; PeFu, *Pelteobagrus fulvidraco*; Rhin, *Rhinogobius*; HeLe, *Hemiculter leucisculus*; CaAu, *Carassius auratus*; Oreo, *Oreochromis*; AcBa, *Acheilognathus barbatulus*; CyCa, *Cyprinus carpio*; CyCh, *Cyprinus chila*; PsPa, *Pseudorasbora parva*; HyNo, *Hypophthalmichthys nobilis*; HyMo, *Hypophthalmichthys molitrix*; Misg, *Misgurnus*; AbRi, *Abbottina rivularis*; MaRo, *Macrobrachium rosenbergii*; MaNi, *Macrobrachium nipponense*; Gast, gastropod; Zoob, zoobenthos; Zoop, zooplankton; Phyt, phytoplankton; Macr, macrophytes; Peri, periphyton; SOM, sedimentary organic matter; Tede, terrestrial detritus.

enrichment pattern, with most basal sources and consumers showing the highest $\delta^{13}C$ and $\delta^{15}N$ values in Qilu Lake and the lowest in Yilong Lake, forming a clear gradient of increasing isotopic enrichment from Yilong to Xingyun to Qilu lakes. The EcoDiet mixing model was used to quantify the dietary contributions of different food sources to consumers and revealed pronounced inter-lake differences in food web structure and trophic interactions (Figure 1B).

In all three lakes, zooplankton were the predominant prey for most consumers, with phytoplankton accounting for 72%–99% of their diet (Figures 1B and 2). The contribution proportions of zoobenthos to certain consumers were relatively higher in Xingyun Lake than in the other two lakes, particularly for species such as *Rhinogobius* (30%), *Hemiculter leucisculus* (25%), and *Misgurnus* (27%). In Yilong Lake, terrestrial detritus made a relatively greater contribution to primary consumers, such as zooplankton (28%) and snails (10%). The major prey of carnivorous fishes differed among lakes. In Yilong Lake, *Culter erythropterus* and *Pelteobagrus fulvidraco* relied primarily on shrimp, whereas in Xingyun and Qilu lakes, fish prey constituted the dominant food source for carnivorous species, contributing most to species such as *Silurus asotus*, *C. erythropterus* and *P. fulvidraco*. SOM and periphyton were the major carbon sources for snail, whereas macrophytes contributed relatively little to consumers in all three lakes (Figure 2; Table S2).

Basal source relative importance and food web stability

Among the five basal resources, phytoplankton was the dominant basal source for consumers in all three lakes (Figure 3). Its relative importance was substantially greater than that of periphyton, macrophytes, SOM, and terrestrial detritus. However, the relative importance of phytoplankton differed among lakes, with the highest contribution in Qilu Lake, followed by Xingyun Lake, and the lowest in Yilong Lake (Table S3). This pattern indicates increasing consumer dependence on phytoplankton-derived carbon sources along the productivity gradient.

Linear mixed-effects models revealed significant relationships between phytoplankton dominance and food web stability. Higher phytoplankton relative importance was associated with lower energy-channel diversity ($\beta = -0.616 \pm 0.005$, $p < 0.001$)

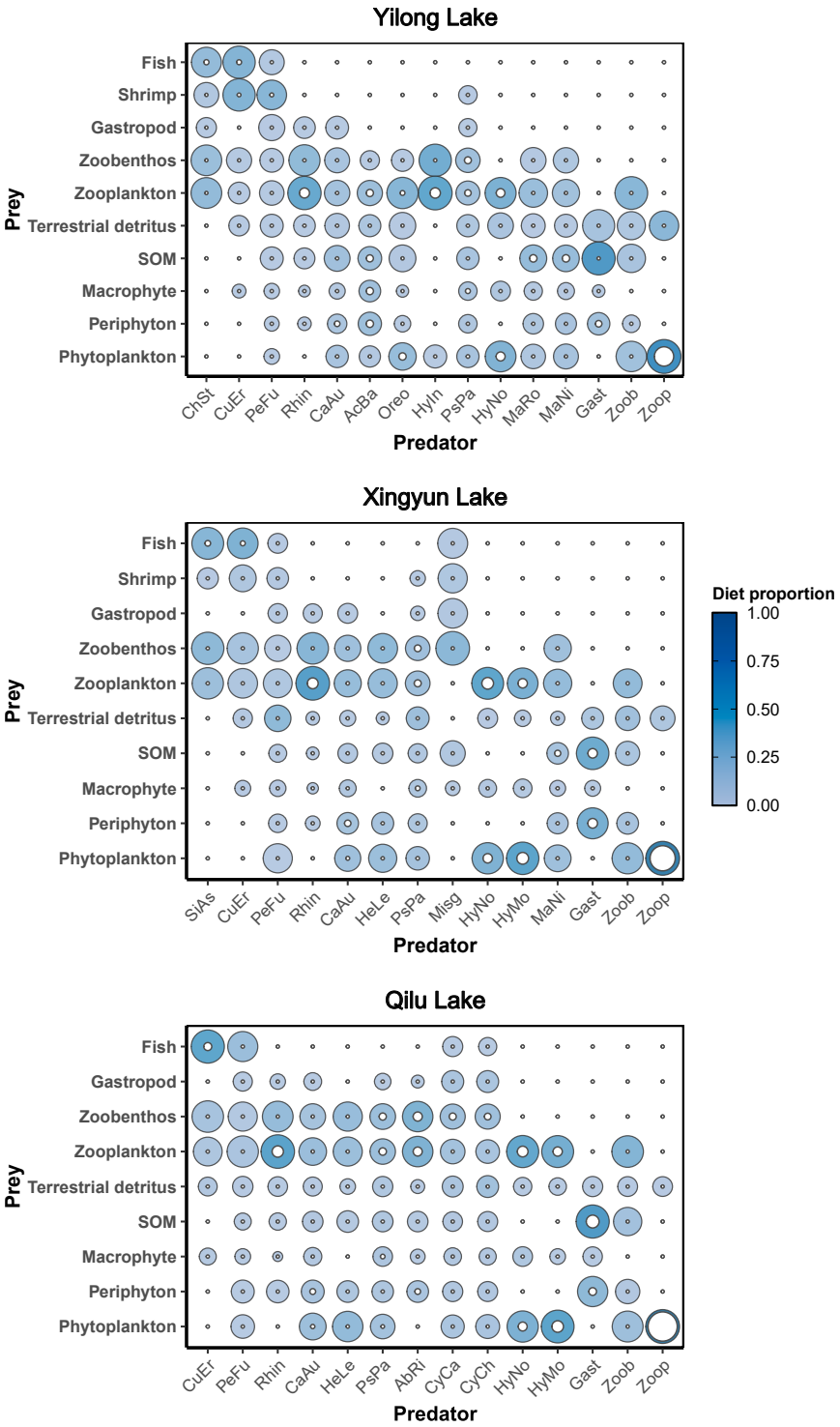


Figure 2. Posterior medians and 90% credible intervals of dietary proportions for each consumer estimated by EcoDiet mixing models

Circle color indicates the posterior median dietary contribution, whereas inner and outer radii represent the 5th and 95th percentiles, respectively. SiAs, *Silurus asotus*; CuEr, *Culter erythropterus*; ChSt, *Channa striata*; HyIn, *Hyporhamphus intermedius*; PeFu, *Pelteobagrus fulvidraco*; Rhin, *Rhinogobius*; HeLe, *Hemiculter leucisculus*; CaAu, *Carassius auratus*; Oreo, *Oreochromis*; AcBa, *Acheilognathus barbatulus*; CyCa, *Cyprinus carpio*; CyCh, *Cyprinus chila*; PsPa, *Pseudorasbora parva*; HyNo, *Hypophthalmichthys nobilis*; HyMo, *Hypophthalmichthys molitrix*; Misg, *Misgurnus*; AbRi, *Abbottina rivularis*; MaRo, *Macrobrachium rosenbergii*; MaNi, *Macrobrachium nipponense*; Gast, gastropod; Zoob, zoobenthos; Zoop, zooplankton; SOM, sedimentary organic matter.

and trophic redundancy ($\beta = -0.422 \pm 0.003$, $p < 0.001$), indicating an increasing reliance on a single, phytoplankton-based energy channel. In contrast, weighted modularity increased with the relative importance of phytoplankton ($\beta = 0.073 \pm 0.004$ SE, $p < 0.001$), suggesting that greater dominance of pelagic trophic pathways corresponded to more compartmentalized food web structure (Figure 4).

DISCUSSION

Our study reveals that eutrophication does not simply intensify algal production, but reconfigures basal energy-channel architecture even within ecosystems characterized as eutrophic. Across the productivity gradient in three shallow plateau lakes, the proportional contributions of phytoplankton to consumer diets increased markedly while reliance on benthic and detrital sources declined. These changes coincided with declines in energy-channel diversity and trophic redundancy, suggesting a structural concentration of basal resource use. At the same time, weighted modularity increased with phytoplankton dominance, which reflects enhanced compartmentalization of trophic interactions. These patterns reflected a structural trade-off under eutrophication: as phytoplankton dominance intensified, food webs shifted toward a more concentrated reliance on a single pelagic pathway, accompanied by greater internal modular organization but reduced cross-channel redundancy. Although elevated modularity may locally constrain disturbance propagation within modules, the overarching dependence on phytoplankton-derived pathway suggested increased vulnerability of whole food webs to fluctuations mediated by phytoplankton dynamics.

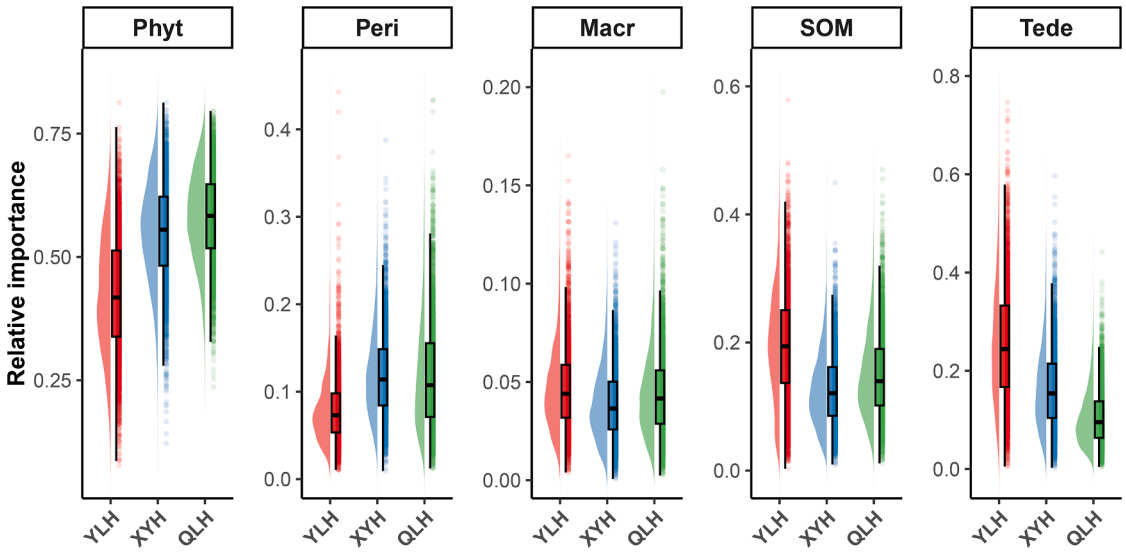


Figure 3. Posterior distribution of the relative importance of five basal sources across three lakes

Phyt, phytoplankton; Peri, periphyton; Macr, macrophytes; SOM, sedimentary organic matter; Tede, terrestrial detritus; YLH, Yilong Lake; XYH, Xingyun Lake; QLH, Qilu Lake.

Eutrophication promotes phytoplankton-dominated energy pathways

Across the three eutrophic shallow lakes, phytoplankton were the dominant basal resource for consumers, with its relative importance increasing along the gradient of eutrophic intensity. This trajectory corresponded with elevated nutrient concentrations (TP) and primary productivity (chlorophyll-a), suggesting that the massive accumulation of algal biomass under eutrophic conditions directly drove trophic reallocation. Rather than supporting by multiple parallel energy channels, increasingly productive conditions favored consumer assimilation of pelagic-derived carbon.[6,13,29–31] Similar shifts toward pelagic dominance under eutrophication have been proved in experimental and theoretical studies, where enhanced algal production often coincides with weakened benthic or macrophyte-based pathways.[18,22,32] Our isotope-based quantification indicated a measurable reorganization of consumer diets toward phytoplankton-derived resources, thereby restructuring the basal architecture of the food web.

Stable isotope signatures provided an empirical fingerprint of this trophic transformation. Along the productivity gradient, both basal sources and consumers displayed a consistent enrichment in $\delta^{13}C$ and $\delta^{15}N$ values, which points to a tightening trophic coupling between phytoplankton-derived production and higher trophic-level consumers.[33] This isotopic convergence suggested that pelagic carbon was increasingly transmitted through multiple consumer groups under eutrophic conditions. Importantly, the isotopic patterns reflected a qualitative shift in assimilation pathways rather than simply greater biomass at the base of the food web.[22,34]

Although nutrient enrichment favored pelagic production across the three lakes, the intensity and persistence of phytoplankton dominance were also modulated by lake-specific contexts, including watershed nutrient inputs, internal nutrient release from sediments, hydrological residence time, and mixing dynamics.[35–37] These interacting processes determine the availability of nutrients as well as their retention and recycling in systems, thereby affecting ecological responses. For instance, the high algal biomass observed in Qilu Lake was associated with surface-concentrated production and potential light limitation,[38] which may reduce benthic primary production. The subsequent deposition and decay of phytoplankton biomass may increase sedimentary oxygen demand, further suppressing benthic communities and associated energy pathways.[39,40] In contrast, Xingyun Lake maintained a relatively more developed benthic pathway, demonstrating that eutrophication does not produce uniform food web structures across lakes. Therefore, phytoplankton dominance should be viewed as an emergent property of interacting external loading and internal biogeochemical and hydrological processes. Such lake-specific contexts could regulate how eutrophication-driven pelagic production is assimilated and propagated through food webs.

Effects of phytoplankton dominance on food web stability

Our results showed that increasing phytoplankton dominance was associated with declines in both basal energy-channel diversity and trophic redundancy across the three lakes. As quantified in our network metrics, consumers in more productive systems derived a larger proportion of their diets from phytoplankton, while the relative contribution of alternative basal resources reduced.[16,25] Energy-channel diversity, which reflects the extent to which multiple basal resources contribute to consumer diets, therefore decreased along the productivity gradient, suggesting a contraction in the range of basal pathways supporting higher trophic levels. Similarly, reduced trophic redundancy indicated lower functional overlap among consumers in their dependence on basal resources, rendering it

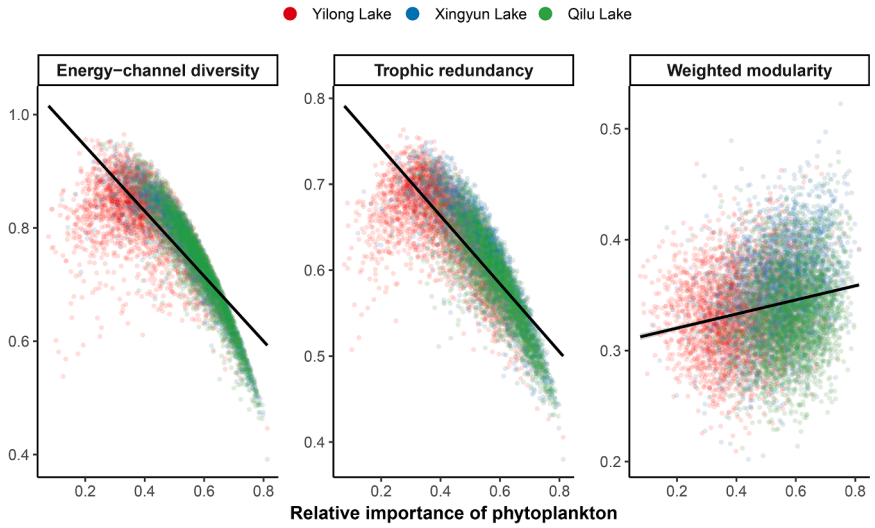


Figure 4. Relationships between the relative importance of phytoplankton and food web stability metrics across three lakes

Points denote posterior samples, while the solid lines represent fitted trends derived from linear mixed-effects models with lake included as a random intercept.

more sensitive to fluctuations in phytoplankton production.[41,42] Moreover, population fluctuations in pivotal consumers (e.g., zooplankton or filter-feeding fish) will be rapidly amplified through these streamlined pathways, which may in turn trigger large-scale oscillations and unstable dynamics throughout the network.[12,43,44] It is important to notice that these metrics represent structural properties inferred from proportional diet contributions rather than direct measurements of dynamic stability, but they provide insight into the potential sensitivity of the food web to basal resource variability.

Conversely, increased phytoplankton dominance corresponded to higher interaction-weighted modularity, partitioning the networks into more relatively independent subgroups with denser internal than external interactions.[45,46] These compartmentalized architectures suggested that phytoplankton-derived carbon was not distributed uniformly across the entire network but was transmitted through relatively discrete trophic guilds,[47] which likely arises from variation in consumer assemblages and prey preferences. Such modular organization may buffer disturbances by containing them locally, thereby preventing network-scale collapse and fostering species persistence within modules.[48–50] However, in the systems we study, such structural compartmentalization occurred alongside increasing dependence on a single basal pathway. Consequently, while modularity may enhance local containment of disturbances, it cannot functionally offset the erosion in energy-channel diversity or trophic redundancy. The coexistence of stronger compartmentalization and weaker cross-channel redundancy highlights that different dimensions of structural organization respond in distinct ways to eutrophication.

These findings manifested a structural trade-off associated with eutrophication in shallow lakes. As phytoplankton dominance intensified, food webs exhibited increased compartmentalization but reduced diversity and redundancy of basal energy channels. By channeling energy through more readily accessible and tightly coupled pelagic pathways, growing reliance on phytoplankton streamlines food web structure,[16,51] a configuration that can boost production and sustain consumer biomass under favorable conditions.[52,53] Yet, this structural organization is underpinned by functional narrowing of basal resource use. The erosion of alternative energy pathways may disrupt long-term stability when food web is confronted with stochastic perturbations, such as harmful algal blooms, nutrient loading pulses, and climate-driven changes in key physicochemical variables.[54,55] Although increased modularity may confer a certain degree of structural resilience, it remains a fragile defense that cannot fully mitigate functional vulnerability imposed by single resource dependence.[17,47] The decoupling between structural compartmentalization and functional resource diversity reveals how eutrophication reshapes food webs. It trades the functional stability provided by diverse pathways for the structural stability of compartmentalization, thereby altering the food web's response to environmental stressors.

Implications for ecological conservation and restoration

The food web reorganization observed along the eutrophication gradient offers pivotal insights for the conservation and restoration of shallow lake ecosystems. The transition toward phytoplankton dominance is conventionally viewed as a shift toward a more vulnerable ecosystem state.[8,10] Our study confirms that eutrophication alters primary productivity while restructuring the basal energy architecture of food web by channeling consumer diets away from diverse basal sources toward a narrower, phytoplankton-based pathway. This structural simplification, evidenced by declining energy-channel diversity and trophic redundancy, indicates reduced functional buffering capacity, even in systems where overall biomass production remains high. Consequently, the resilience of these lakes becomes increasingly precarious. Future management interventions therefore need to extend beyond the control of nutrient loads and algal blooms. They should synchronously safeguard the structural and functional integrity of the food web. Protecting and restoring diverse basal carbon pathways, such as improving benthic habitats, promoting submerged macrophyte recovery, and reducing sediment disturbance, may help restore benthic and detrital pathways that diversify energy flow, which is essential for sustaining trophic redundancy that underpins ecosystem resilience against environmental perturbations.[56]

The concomitant rise in weighted modularity characterizes the food web's responds to disturbances, but these systems remain particularly susceptible to environmental fluctuations that directly destabilize phytoplankton dynamics, including extreme climate events, altered mixing regimes, or variations in nutrient

stoichiometry.[57–59] Ultimately, restoration efforts should pay attention to restore the functional complexity at the base of food webs and foster a coordinated balance among productivity, energy channel diversity, and network structure.

Limitations of the study

Given that all three plateau shallow lakes were eutrophic, the inferred "gradient" reflects differences in trophic intensity (e.g., chlorophyll-a, TP, TN, and phytoplankton biomass) rather than a transition across different trophic states. Hence, our findings are confined to eutrophic systems and warrant caution when extrapolating to oligotrophic or mesotrophic conditions. In addition, this study represents a temporally constrained snapshot, as sampling was conducted at representative sites within each lake during a single campaign. Consequently, it cannot fully resolve fine-scale spatial heterogeneity or long-term temporal dynamics.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Jun Xu (xujun@ihb.ac.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All data reported in this paper will be shared by the lead contact upon request.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, J.-Y.X., P.-Y.Z., and J.X.; methodology, J.-Y.X., P.-Y.Z., T.W., and J.X.; investigation, J.-Y.X., X.-F.N., L.-W.X., and Y.-Z.H.; writing – original draft, J.-Y.X.; writing – review and editing, J.-Y.X., P.-Y.Z., T.W., X.-F.N., H.Z., G.-H.S., and J.X.; funding acquisition, J.X., P.-Y.Z., T.W., and H.Z.; supervision, P.-Y.Z., H.Z., G.-H.S., and J.X.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological samples		
water samples	Yilong Lake, Xingyun Lake, Qilu Lake in Yunnan Province	N/A
sediment samples	Yilong Lake, Xingyun Lake, Qilu Lake in Yunnan Province	N/A
fish, shrimp and snail tissue	Yilong Lake, Xingyun Lake, Qilu Lake in Yunnan Province	N/A
plant leaves	Yilong Lake, Xingyun Lake, Qilu Lake in Yunnan Province	N/A
Software and algorithms		
R	version 4.4.1	https://www.r-project.org

METHOD DETAILS

Study area

Yilong Lake, Xingyun Lake and Qilu Lake are located in central Yunnan (23.67–24.37 °N), with mean water depths ranging from approximately 2.4 to 5.3 m and maximum depths generally below 11 m.[60] Featuring closed or semi-closed basin morphologies, these shallow lakes exhibit limited thermal stratification and undergo continuous water mixing. Prolonged anthropogenic nutrient loading and pollutant accumulation from the surroundings have consequently diminished their resistance to external disturbances and capacity for self-recovery.[61] While the three lakes share similar characteristics in geographic location, surface area and lake type, their watershed characteristics, hydrological regimes and ecological conditions are different. Over recent decades, all three lakes have suffered varying degrees of eutrophication, thus offering a natural comparative case to assess how food web structure and trophic pathways vary along a productivity gradient.

In this study, the productivity gradient refers to differences in nutrient concentrations and algal biomass among the three lakes within the eutrophic range. We define the gradient based on multiple indicators of trophic state, including chlorophyll-a concentration, TP, TN and phytoplankton biomass.

Sampling and processing

Field sampling was conducted in May 2025 at six sites in each lake (Figure S2). Water quality parameters including water temperature, conductivity, pH and dissolved oxygen were measured *in situ* using a portable meter (HQ60d, HACH, USA). We collected 1 L water samples for nutrient and chlorophyll-a analysis. TN and TP concentrations were determined by spectrophotometry after alkaline potassium persulfate digestion of unfiltered water samples. Chlorophyll-a was extracted in acetone from samples filtered onto 0.45 μm Whatman GF/C filters and subsequently measured by spectrophotometry.

Integrated water samples from the surface (0.5 m below the water surface) and bottom layers (0.5 m above the sediment) were collected using a Plexiglass water sampler. For phytoplankton analysis, a 1 L subsample was fixed with Lugol's iodine solution. Zooplankton (copepods and cladocerans) were collected by filtering 10 L of the integrated water through a 112-μm plankton net, while rotifers were enumerated from the same phytoplankton samples. Zoobenthos were quantitatively sampled using a Peterson grab (1/16 m^2) and then washed through a hand net to separate organisms from the sediment. All organisms were identified to the lowest possible taxonomic level, and their biomasses were determined.[62–64]

To determine changes in food web structure and differences in dietary contributions along the productivity gradient, we collected stable isotope samples of major trophic groups from three lakes, including phytoplankton, periphyton, submerged macrophytes, SOM and terrestrial detritus as potential basal carbon sources, as well as fish, shrimp, snails, zoobenthos and zooplankton as representative consumers.

Zooplankton were collected using a 112-μm mesh plankton net, and the filtered water was then passed through a 64-μm mesh net to obtain phytoplankton samples. The collection of zoobenthos followed the same procedure as that used for quantitative sampling, while SOM samples were simultaneously collected from the top 1 cm of the sediment surface. Benthic invertebrates were preserved in distilled water for 12 h to clear their gut contents,[65] and all individuals from the same site were pooled to obtain a sufficient composite sample. Fish and shrimp were captured using multi-panel gillnets and fish cages. For stable isotope analysis, dorsal muscle tissue from fish (after skin removal) and muscle tissue from shrimp (after removing the exoskeleton) were dissected and separately stored in 2 mL centrifuge tubes. We collected submerged macrophytes using an aquatic plant sampler in nearshore areas adjacent to the sampling sites, while shed leaves of dominant shoreline grassland plants were gathered as potential terrestrial detritus. Both plant materials were rinsed with distilled water to remove surface debris particles and attached organisms. Concurrently, periphyton was brushed from stone surface along the shoreline using a toothbrush and transferred into bottles containing a small amount of distilled

water. Snails were collected from the same area, and the soft tissue was carefully dissected from the shells and preserved in centrifuge tubes for subsequent analysis.

All stable isotope samples were transported to the laboratory after collection for processing. Zooplankton, phytoplankton and periphyton samples were filtered onto pre-combusted (450°C for 4 h) Whatman GF/C filters (0.45 μm). To minimize contamination, particulate contaminants were manually picked out from samples. Additionally, for phytoplankton samples, zooplankton individuals were carefully removed to avoid isotopic interference. Following drying to constant weight at 60°C (at least 48 h), all samples were homogenized into fine powder using a bead mill homogenizer (MiniBeadbeater-16, Glen Mills, Inc., California, USA), and then packed into tin capsules for mass spectrometric analysis.

Stable isotope analyses and modeling

Carbon and nitrogen stable isotope ratios ($\delta^{13}C$ and $\delta^{15}N$) were determined at the Institute of Hydrobiology, Chinese Academy of Sciences using a Delta Plus (Finnigan, Bremen, Germany) isotope ratio mass spectrometer coupled with a Carlo Erba NC 2500 elemental analyzer (Carlo Erba Reagenti, Milan, Italy). Stable isotope ratios are expressed in delta (δ) notation in parts per thousand (‰) relative to international standards, according to the following equation:

$$\delta X\ (‰) = \left[\left(\frac{R_{sample}}{R_{standard}}\right) - 1\right] \times 1000$$

where X represents ^{13}C or ^{15}N, and R is the corresponding ratio of $^{13}C/^{12}C$ or $^{15}N/^{14}N$. The reference standard for carbon was Vienna Pee Dee Belemnite (VPDB), and that for nitrogen was atmospheric nitrogen (N_2). Analytical precision was less than 0.2‰ for $\delta^{13}C$ and 0.3‰ for $\delta^{15}N$.

Food web nodes were defined based on ecologically meaningful trophic units. Basal resources (phytoplankton, periphyton, submerged macrophytes, SOM and terrestrial detritus) were treated as independent nodes representing distinct basal energy pathways. Zooplankton, snails and zoobenthos (primarily oligochaetes and chironomids) were aggregated into single trophic groups, functioning as key intermediate consumers with similar trophic roles. Fish and shrimp were resolved at the species level because they occupy distinct trophic positions and often exhibit species-specific feeding strategies. Predator-prey interactions were determined based on published dietary information (Table S4).[66–68] A binary diet matrix specifying potential trophic links was constructed and used as structural input in the Bayesian mixing model EcoDiet,[69] which was combined with carbon and nitrogen stable isotope values to estimate diet proportions for each predator. To account for isotope fractionation during the process of digestion and metabolism, the trophic enrichment factors of 0.8‰ for $\delta^{13}C$ and 3.4‰ for $\delta^{15}N$ were applied to correct the stable isotope values of food sources in the model.[70,71] We aggregated prey taxa into ten trophic groups to highlight the major trophic pathways. For each Markov chain Monte Carlo (MCMC) iteration (3000 posterior samples for each lake), the estimated dietary contributions of prey belonging to the same group were summed.

Quantification of relative importance and stability-related metrics

The dietary contributions of each basal carbon source were propagated through the food web to quantify their assimilation by each consumer. Within each MCMC iteration, this was achieved by multiplying basal source-to-prey contributions by prey-to-predator contributions, and then aggregating the results across all pathways. To ascertain the relative importance of individual basal resource, we summed its posterior contributions across the entire consumer assemblage, normalized by the total contributions of all basal sources. This indicator reflects the aggregate demand for basal resource in the food web by integrating the diet compositions of all trophic-level consumers into a composite metric, and it allows for the comparison of resource dependence across different lakes. It is important to note that these estimates represent proportional dietary assimilation rather than absolute energy fluxes. Consequently, identical proportional contributions do not imply equivalent magnitudes of energy transfer, as consumers differ in biomass, metabolic demand, ingestion rates and turnover.[72] Our inference focuses on the structural allocation of basal energy pathways within the food web rather than on quantifying ecosystem-scale energy flux.

We adopted a structural perspective and evaluated stability-related properties using three complementary network metrics, including basal energy-channel diversity, trophic redundancy and interaction-weighted modularity to consider how reliance on phytoplankton as a basal carbon source may affect food web stability. Energy-channel diversity was quantified by normalizing the Shannon entropy of the relative importance of the five basal sources:

$$H_{EC}^{norm} = \frac{-\sum_{k=1}^{K} p_k \ln(p_k)}{\ln(K)}$$

where H_{EC}^{norm} denotes the normalized energy-channel diversity, P_k is the relative importance of the k-th basal resource, and K is the total number of basal resources (here $K = 5$). This normalization constrains H_{EC}^{norm} to the range [0, 1]. Higher values indicate a more balanced utilization of multiple basal pathways by consumers, which distributes trophic support and may enhance food web stability by reducing dependence on any single resource.

Trophic redundancy, defined as the complement to Simpson's diversity index of basal source contributions to each consumer, reflects the degree of dietary overlap supported by multiple basal sources. For an individual consumer j, trophic redundancy was calculated as:

$$R_j = 1 - \sum_{i=1}^{S_b} p_{ij}^2$$

where p_{ij} represents the relative contribution of basal resource i to consumer j, and S_b is the number of basal resources (here $S_b = 5$). Network-level trophic redundancy was then calculated as the posterior mean across all consumers:

$$R_{web} = \frac{1}{N} \sum_{j=1}^{N} R_j$$

where N is the total number of consumers in the food web.

Interaction-weighted modularity was employed to capture the heterogeneity in interaction strengths,[73] where edge weights represent the posterior medium dietary contribution proportions between consumers and their prey derived from the EcoDiet model, providing a basis for evaluating food web compartmentalization. The calculation formula is as follows:

$$Q_w = \frac{1}{2W} \sum_{ij} \left(w_{ij} - \frac{s_i s_j}{2W} \right) \delta(c_i, c_j)$$

where w_{ij} is the interaction strength from species i to species j, $s_i = \sum_j w_{ij}$ is the strength of node i, and $W = \sum_{ij} w_{ij}$ is the total interaction weight in the network. The term c_i denotes the module assignment of node i, and $\delta(c_i, c_j)$ is an indicator function equal to 1 when nodes i and j belong to the same module and 0 otherwise. Higher values of Q_w indicate a more modular organization of the food web network.

These metrics capture distinct mechanisms linked to ecological stability, including resource pathway diversification, functional buffering capacity and disturbance containment through compartmentalization. Refer to Text S1 for the detailed equations for each metric. Results were presented as posterior medians with 90% credible intervals (5th-95th percentiles) derived from Bayesian posterior distributions.

QUANTIFICATION AND STATISTICAL ANALYSIS

One-way ANOVA was used to determine differences in water quality parameters and biomasses, supplemented by Tukey's HSD post-hoc testes for pairwise comparisons between lakes. Prior to analysis, model assumptions of normality and homoscedasticity were evaluated via Shapiro-Wilk and Levene's tests, respectively. If the assumptions were not satisfied, response variables were log- or square-root-transformed to improve conformity with model requirements. We used linear regression to examine the relationship of phytoplankton biomass against TN and TP concentrations. To assess how the relative importance of phytoplankton relates to each stability metric, we fitted linear mixed-effects models (LMMs). The phytoplankton relative importance was the fixed effect, and lake was included as a random intercept. Analyses were conducted using the 'lmer' function from the *lme4* package. All statistical analyses, modeling and visualizations were performed in R (version 4.4.1).